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| <b>DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE</b><br><b>Supplementary Examination – Summer 2022</b><br><b>Course: B. Tech.SE                      Branch :CSE                      Semester :-III</b><br><b>Subject Code &amp; Name:Computer Architecture &amp; Organization (BTCOC304)</b><br><b>Max Marks: 60                      Date:                      Duration: 3 Hr.</b> |                                                                                                |            |            |
| <b>Instructions to the Students:</b><br>1. All the questions are compulsory.<br>2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.<br>3. Use of non-programmable scientific calculators is allowed.<br>4. Assume suitable data wherever necessary and mention it clearly.               |                                                                                                |            |            |
|                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                | (Level/CO) | Marks      |
| <b>Q. 1</b>                                                                                                                                                                                                                                                                                                                                                                                      | <b>Solve Any Two of the following. (This is just a sample instruction)</b>                     |            |            |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                                        | <b>Write a note on ALU design.</b>                                                             |            | <b>06</b>  |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                                        | <b>Write a note on computer components.</b>                                                    |            | <b>06</b>  |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                                                                        | <b>Define terms: Single-cycle and multi-cycle Implementation</b>                               |            | <b>06</b>  |
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| <b>Q.2</b>                                                                                                                                                                                                                                                                                                                                                                                       | <b>Solve Any Two of the following. (This is just a sample instruction)</b>                     |            |            |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                                        | <b>Explain RISC and CISC architecture.</b>                                                     |            | <b>06</b>  |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                                        | <b>Draw and explain single-cycle datapath for the memory instructions,</b>                     |            | <b>06</b>  |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                                                                        | <b>Explain in detail elements of Bus design.</b>                                               |            | <b>06</b>  |
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| <b>Q. 3</b>                                                                                                                                                                                                                                                                                                                                                                                      | <b>Solve Any One of the following. (This is just a sample instruction)</b>                     |            |            |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                                        | <b>Explain the following addressing modes: i) Immediate ii) Indirect<br/>iii) Displacement</b> |            | <b>06</b>  |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                                        | <b>Write a short note on Disk Storage.</b>                                                     |            | <b>06</b>  |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                                                                        | <b>Draw and explain in detail the concept of virtual memory with page fault.</b>               |            | <b>06</b>  |
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| <b>Q.4</b>                                                                                                                                                                                                                                                                                                                                                                                       | <b>Solve Any Two of the following. (This is just a sample instruction)</b>                     |            |            |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                                        | <b>Draw and explain RAID levels in detail.</b>                                                 |            | <b>006</b> |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                                        | <b>Write a note on the Hardwired Control unit.</b>                                             |            | <b>06</b>  |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                                                                        | <b>Write a note on Magnetic tape and Optical Memory.</b>                                       |            | <b>06</b>  |
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| <b>Q. 5</b>                                                                                                                                                                                                                                                                                                                                                                                      | <b>Solve Any One of the following. (This is just a sample instruction)</b>                     |            |            |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                                        | <b>Define and explain in detail Data Hazard with examples.</b>                                 |            | <b>06</b>  |

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| B) | Draw and explain in detail Memory Hierarchy. Difference between main memory and cache memory. |  | 06 |
| C) | What is DMA? Draw typical DMA module diagram.                                                 |  | 06 |
|    | *** End ***                                                                                   |  |    |

The grid and the borders of the table will be hidden before final printing.